Notes on Radio Evolution of Galaxies

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Contents

Chapter 1

Useful Terminology and Concepts

1.1 Basic Terminology

i. add parsec defn

ii. **Luminosity** (L): The total amount of electromagnetic **energy emitted per unit time** by an astronomical object.

iii. **Radiant Flux Density** (S): The total amount of electromagnetic **energy per unit area per unit time** by an astronomical object. In other words, it is the luminosity per unit area. According to this definition, at a distance r from a star (say):

$$S = \frac{L}{4\pi r^2}$$

iv. **Effective Temperature**: The effective temperature of a body such as a star or planet is the temperature of a black body that would emit the same total energy as electromagnetic radiation.

v. **Specific Luminosity** (L_ν): The energy emitted per unit time for a unit frequency range.

1.2 The Magnitude Scale

Our eyes judge brightness according in a logarithmic fashion, similar to hearing. This means that what we perceive to be twice as bright is actually emitting a lot more than just twice the radiant energy. Due to this, optical astronomy uses 'magnitude' as a scale of brightness.

1.2.1 Apparent Magnitude

The purpose of apparent magnitude is to closely match visually reported magnitudes and quantify these historical observations. It is a measure for the radiant flux S from a body. For two sources with fluxes S_1 and S_2, their *apparent magnitudes* m_1 and m_2 are defined as:

$$m_1 - m_2 = -2.5 \log\left(\frac{S_1}{S_2}\right)$$

This means that a brighter object will have a smaller apparent magnitude. The factor of 2.5 is chosen to most accurately match the historically calculated magnitudes. A more *absolute* definition can be found below in the next section, which depends on the filter in consideration.

1.2.2 Filters and Colour

Optical observations are performed with a combination of filters and detectors. Since the spectral energy distribution is not necessarily uniform over all wavelengths, apparent magnitudes are defined for each of these filters, i.e. apparent magnitude is a function of the wavelength of light being considered.

Vega Magnitude

$$m = m(\lambda) \equiv m_X \equiv X$$

where X is the filter for some specific wavelength λ.

The magnitudes of the different filters are related, by definition, as $U = B = V = R = I = \ldots$ for main sequence stars of spectral type A0 (Vega is an archetype of such stars), i.e., every colour index for such a star is defined to be zero.

For a more precise definition, let $T_X(\nu)$ be the transmission curve of the filter-detector system. $T_X(\nu)$ specifies which fraction of the incoming photons with frequency are registered by the detector. The apparent magnitude of a source with spectral flux S_ν is then:

$$m_X = -2.5 \log\left(\frac{\int \mathrm{d}\nu \; T_X(\nu) S_\nu}{\int \mathrm{d}\nu \; T_X(\nu)}\right) + \text{const.}$$

The argument of the logarithm is the average flux received in total for the entire EM spectrum. The constant is determined from reference stars. This is also referred to as the Vega apparent magnitude.

AB Magnitude

Another commonly used definition, does not consider the spectral energy distribution of reference. The reference is instead defined as constant flux at all frequencies, $S_\nu^{\mathrm{ref}} = S_\nu^{\mathrm{AB}} = 2.89 \times 10^{-21}$ erg s^{-1} cm^{-2} Hz^{-1}. This value makes A0 stars like Vega have the same magnitude in the original Johnson-V-band as in the AB system.

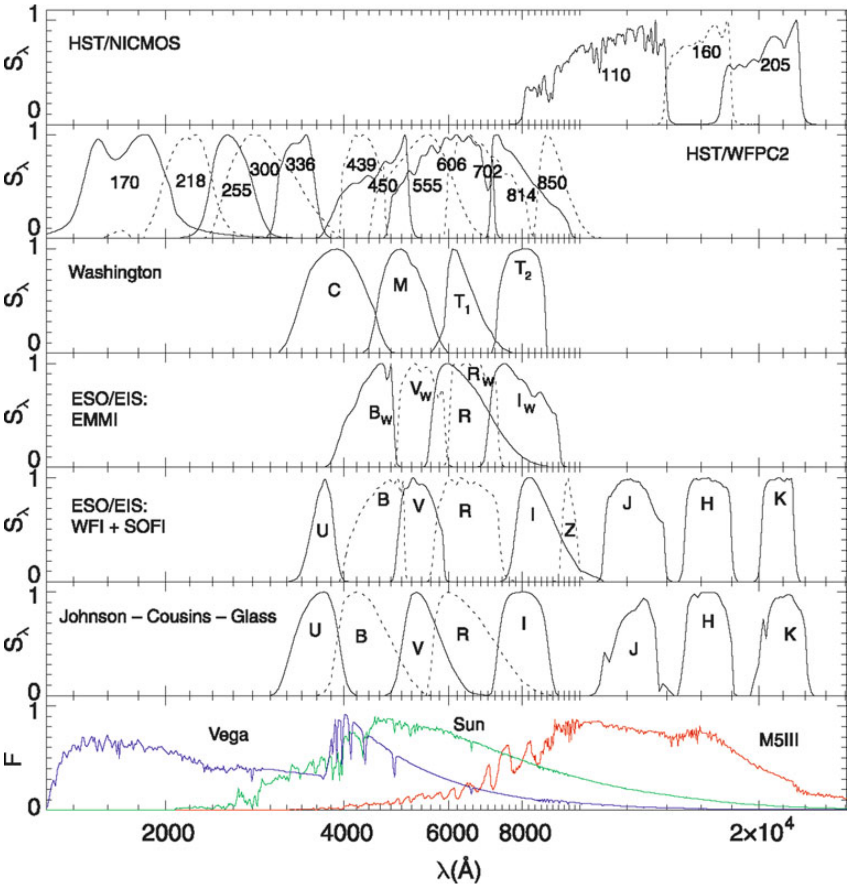

Figure 1.1: This shows the transmission curves of different spectra. At the bottom, we have the spectra of 3 stars with different effective temperatures.

1.2.3 Absolute Magnitude

Since apparent magnitude does not reveal any information about the luminosity of the body since it inherently depends on the distance of the body from Earth. If we are at a distance D from a body and L_ν (energy per unit time for unit frequency range) is its specific luminosity, the flux for isotropic emission is given by:

$$S_\nu = \frac{L_\nu}{4\pi D^2}$$

To reveal the physical properties of the source itself irrespective of how it appears to us, we define absolute magnitude, M_X, where X refers to the filter under consideration. It is defined to be the apparent magnitude of a body at a distance of 10 pc from us. Its relation to the apparent magnitude is as follows:

$$m_X - M_X = 5\log\left(\frac{D}{1pc}\right) - 5 \equiv \mu$$

where μ is called the *distance modulus*. Hence, the latter is a logarithmic measure of the distance of a source: $\mu = 0$ for $D = 10$ pc, $\mu = 10$ for $D = 1$ kpc, and $\mu = 25$ for $D = 1$ Mpc. The difference between apparent and absolute magnitude is independent of the filter choice, and it equals the distance modulus if no extinction is present. In general, this difference is modified by the wavelength (and thus filter) dependent extinction coefficient.

1.2.4 Bolometric Parameters

The total flux S and luminosity L over all frequencies are naturally be defined as:

$$S = \int_0^\infty S_\nu \, \mathrm{d}\nu; \qquad L = \int_0^\infty L_\nu \, \mathrm{d}\nu$$

These define the *bolometric magnitudes*:

$$\begin{aligned} m_{\text{bol}} &= -2.5\log(S) + \text{const.} \\ M_{\text{bol}} &= -2.5\log(L) + \text{const.} \end{aligned}$$

The constants can be determined from reference stars. In particular, using the sun as reference with $m_{\odot\text{bol}} = -26.83$ and the distance of 1 a.u. gives $\mu = -31.47$, thus giving us:

$$M_{\odot\text{bol}} = m_{\odot\text{bol}} - \mu = 4.74$$

From this, we obtain:

$$M_{\text{bol}} = 4.74 - 2.5\log\left(\frac{L}{L_\odot}\right)$$

where the luminosity of the sun is $L_\odot = 3.85 \times 10^{33}$ erg s^{-1}.

Chapter 2

The World of Galaxies

Ever since the Hubble Deep Field image, the whole of humanity has set their sight to discover what lies far beyond they could ever comprehend. When dealing with extragalactic astronomy, the most notable entities are arguably other galaxies, and this is a chapter all about them.

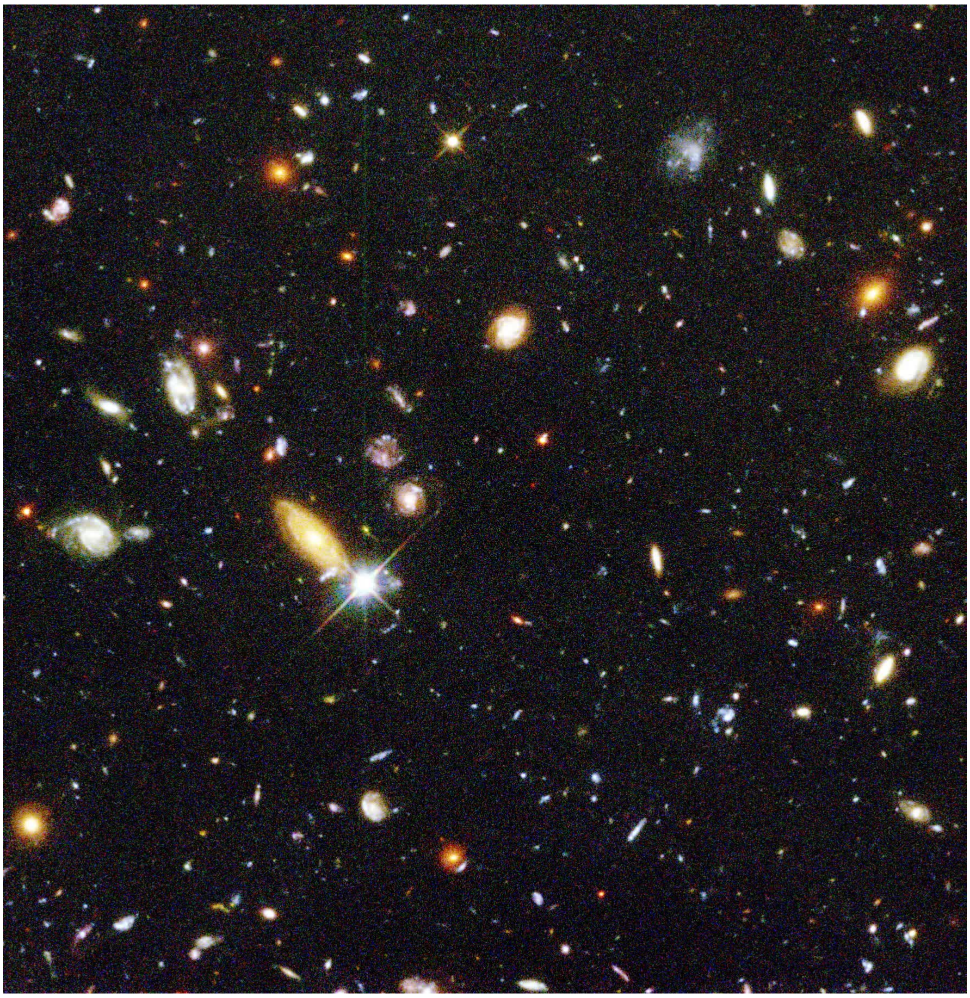

Figure 2.1: The Hubble Deep Field image produced by *Edwin Hubble* changed the world forever. All the dots in the image are galaxies, not stars.

The light from 'normal galaxies' is mainly the sum of all the light emitted by the stars present within it. need to add note about age of stars anol but for that I need to add the section about stars at the beginning.

2.1 Classification of Galaxies

Galaxies dafds

2.1.1 Morphological Classification

On the basis of shape or morphology, galaxies have been historically classified into the following categories:

1. **Elliptical Galaxies**: Their isophotes[1] resemble ellipses. They are further subclassified according to the **ellipticity** of the isophotes: $\epsilon = 1 - b/a$, where a, b are the semi-major and semi-minor axes. Then we define $N = 10\epsilon$ which gives us the representation EN for elliptical galaxies. An E7 galaxy is an elliptical galaxy with $b/a = 0.3$.
2. **Spiral Galaxies**: Their isophotes look like spirals.

[1] Isophotes are contours along which the surface brightness is constant.